



Figure 1: The Dell EMC VxRail hyper-converged infrastructure appliance with Intel Xeon processors  
(Courtesy: Dell EMC - A few quick facts about HCI)

up the latest updates:

1. Nutanix, a leader in HCI, states that the technology streamlines deployment, management and scaling of data centre resources by combining x86-based server and storage resources with intelligent software in a turnkey, software-defined solution.
2. HCI combines compute, storage and the network together as a single appliance. It can be scaled out or expanded using additional nodes. That is, instead of scaling up the traditional way by adding more drives, memory or CPUs to the base system, you scale out by adding more nodes. Groups of such nodes are known as clusters.
3. Each node runs a hypervisor like Nutanix Acropolis, VMware ESXi, Microsoft Hyper-V or Kernel-based Virtual Machine (KVM), and the HCI control features run as a separate virtual machine on every node, forming a fully distributed fabric that can scale resources with the addition of new nodes.
4. Since most modern HCI are entirely software-defined, there is no dependence on proprietary hardware.
5. HCI does not need a separate team as it can be managed by any IT professional. This makes it ideal for small and medium enterprises.
6. HCI is different from the traditional

server and shared-storage architectures. It also differs from the previous generation of converged infrastructure.

7. HCI is not the same as cloud computing!
8. Scale-out and shared-core are two of the keywords you are bound to encounter when reading about HCI. Basically, most HCI implementations involve multiple servers or nodes in a cluster, with the storage resources distributed across the nodes. This provides resilience against any component or node failure. Plus, by placing storage at the nodes, the data is closer to compute than a traditional storage area network. So, you can actually get the most out of faster storage technologies like the non-volatile memory express (NVMe) or non-volatile dual in-line memory module (NVDIMM). The scale-out architecture also enables the company to add nodes as and when required rather than investing in everything upfront. Just add a node and connect it to the network, and you are ready to go. The resources are automatically rebalanced and ready to use. A lot of hyper-converged implementations also have a shared core, that is, the storage and virtual machines compete for the same processors

and memory. This reduces wastage and optimises resource usage.

However, some experts feel that in special cases, the users might have to buy more equipment to run the same workloads.

9. It's true that HCI is great for small enterprises as it can be used without fuss and hassles, but it can be used by really large data centres too. Leading companies have on record hyper-convergence case studies where more than a thousand HCI nodes are installed in a single data centre.
10. In a recent survey, research leader Forrester found that the most common workloads being run on hyper-converged systems are: databases, such as Oracle or SQL server (cited by 50 per cent); file and print services (40 per cent); collaborations, such as Exchange or SharePoint (38 per cent); virtual desktops (34 per cent); commercial packaged software such as SAP and Oracle (33 per cent); analytics (25 per cent); and Web-facing workloads such as the LAMP stack or Web servers (17 per cent).
11. During a recent talk, Dell EMC's European chief technology officer for converged and hyper-converged infrastructure, Nigel Moulton, said that "...mission-critical applications live best in converged infrastructures." He justified his view by saying that many mission-critical applications such as ERP are architected to rely on the underlying hardware (in the storage area network) for encryption, replication and high availability. With HCI, on the other hand, many of these things come from the software stack. Instead of looking at the move to HCI as all-or-none, he suggested that organisations take a practical approach to it. If an application can work on HCI, companies must make the move as and when their existing hardware reaches the end of life. The rest can remain on traditional hardware or converged